

1 Solution — GIAM 2.2

Problem 3

1.1 Problem

Complete a truth table for $A \Rightarrow (B \Rightarrow C)$ and for $(A \Rightarrow B) \Rightarrow C$. What can you conclude about conditionals and the associative property?

1.2 Answer

Conditionals are **not associative** : $A \Rightarrow (B \Rightarrow C) \not\equiv (A \Rightarrow B) \Rightarrow C$.

The two groupings agree on 6 of the 8 rows but differ on the two rows where $A = 0$ and $C = 0$. So the parentheses in a chain of implications genuinely matter and cannot be dropped.

1.3 The Truth Table

Building both formulas over all 8 rows (helper columns $B \Rightarrow C$ and $A \Rightarrow B$):

A	B	C	$B \Rightarrow C$	$A \Rightarrow (B \Rightarrow C)$	$A \Rightarrow B$	$(A \Rightarrow B) \Rightarrow C$
1	1	1	1	1	1	1
1	1	0	0	0	1	0
1	0	1	1	1	0	1
1	0	0	1	1	0	1
0	1	1	1	1	1	1
0	1	0	0	1	1	0
0	0	1	1	1	1	1
0	0	0	1	1	1	1

0	0	0	1	1	1	0
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Table 1.1

The two boldfaced rows are where the final columns disagree.

Both groupings over all 8 rows. The two final columns disagree in the shaded rows — so $\Rightarrow$ is not associative.

A	B	C	$B \Rightarrow C$	$A \Rightarrow (B \Rightarrow C)$	$A \Rightarrow B$	$(A \Rightarrow B) \Rightarrow C$
1	1	1	1	1	1	1
1	1	0	0	0	1	0
1	0	1	1	1	0	1
1	0	0	1	1	0	1
0	1	1	1	1	1	1
0	1	0	0	1	1	0
0	0	1	1	1	1	1
0	0	0	1	1	1	0

Rows differ when $A = 0$ and $C = 0$: the left is vacuously TRUE (A false), but the right collapses to $C = 0$.

Different columns $\Rightarrow A \Rightarrow (B \Rightarrow C)$ is NOT equivalent to $(A \Rightarrow B) \Rightarrow C$.

Implication is not associative — the parentheses cannot be dropped.

Figure 1.1: A combined truth table with columns A, B, C, B implies C, A implies (B implies C), A implies B, and (A implies B) implies C, over all 8 rows. The two output columns are equal in six rows but differ in two shaded rows: $(A,B,C) = (0,1,0)$ and $(0,0,0)$. In both, $A = 0$ so the left formula A implies (B implies C) is vacuously true (value 1), while the right formula (A implies B) implies C has A implies B true and reduces to C, which is 0. Conclusion: implication is not associative; the parentheses cannot be dropped.

1.4 Where and Why They Differ

The two formulas part ways exactly when $A = 0$ and $C = 0$ (rows $(0,1,0)$ and $(0,0,0)$):

- $A \Rightarrow (B \Rightarrow C)$ is **true** there. Its hypothesis A is false, so the whole conditional is *vacuously true* regardless of the inside.
- $(A \Rightarrow B) \Rightarrow C$ is **false** there. With A false, the inner $A \Rightarrow B$ is *true*, so the outer conditional reduces to **true** $\Rightarrow C$, which is just C — and $C = 0$.

So the disagreement comes from *where the false A lands*. On the left A is the outermost hypothesis and short-circuits the sentence to true; on the right A is buried in the antecedent, which a false A makes *true*, exposing C as the deciding value.

1.5 Remark — Contrast With Problem 1: Chained vs. Nested

Beware a tempting confusion with Problem 1. There, “transitivity” chained *separate* implications: from $A \Rightarrow B$ and $B \Rightarrow C$ we got $A \Rightarrow C$. That is a statement about combining two premises, and it is valid. **This** problem is different: it *nests* a single expression and asks whether the grouping can be shifted. It cannot. Transitivity (“ $A \Rightarrow B$ and $B \Rightarrow C$ give $A \Rightarrow C$ ”) is true; associativity (“ $A \Rightarrow (B \Rightarrow C)$ equals $(A \Rightarrow B) \Rightarrow C$ ”) is false. The two are easy to conflate but say entirely different things.

1.6 Remark — The Right-Associative Convention

Because the grouping matters, mathematicians adopt a **convention**: an unparenthesised chain $A \Rightarrow B \Rightarrow C$ is read **right-associatively**, i.e. as $A \Rightarrow (B \Rightarrow C)$. This choice is not arbitrary — the right-nested form is the more useful one, since (via Problem 2’s re-write) it equals

$$A \Rightarrow (B \Rightarrow C) \equiv \neg A \vee (\neg B \vee C) \equiv (A \wedge B) \Rightarrow C,$$

the natural “if A and B , then C ” (a rule with two hypotheses). This equivalence — called **exportation** — is why the right-nested conditional appears constantly in mathematics, while the left-nested $(A \Rightarrow B) \Rightarrow C$ is comparatively rare and, as we saw, behaves quite differently.

1.7 Remark — Which Connectives *Are* Associative

This problem is a companion to Problem 2.1.4, where $\wedge$ and $\vee$ failed to associate *with each other*. The pattern of what associates is worth collecting:

- $\wedge$ associates with itself: $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$. ✓
- $\vee$ associates with itself: $(A \vee B) \vee C \equiv A \vee (B \vee C)$. ✓

- $\Leftrightarrow$ associates: $(A \Leftrightarrow B) \Leftrightarrow C \equiv A \Leftrightarrow (B \Leftrightarrow C)$ (both equal “an even number of A, B, C are false”). ✓
- $\Rightarrow$ does **not** associate (this problem). ✗

So associativity is a special privilege, not a default. The symmetric connectives ($\wedge, \vee, \Leftrightarrow$) enjoy it; the directional $\Rightarrow$, which distinguishes hypothesis from conclusion, does not — its asymmetry is exactly what the two differing rows expose.

1.8 Remark — A Faster Way to See the Non-Equivalence

One need not fill all eight rows to answer the question — a single well-chosen row suffices to *disprove* equivalence. Try to make the two sides differ by killing A and C : set $A = C = \text{false}$ (leave B anything). Then $A \Rightarrow (B \Rightarrow C)$ is true (false hypothesis), while $(A \Rightarrow B) \Rightarrow C$ has a true antecedent $A \Rightarrow B$ and false consequent C , hence is false. One row, $(0, \cdot, 0)$, already settles it. Producing a single **counterexample** row is the standard efficient technique for showing two formulas are *not* equivalent — you only need the full table to show that they *are*.